



HACKSPHERE 2026 PHASE 1 PROPOSAL IDEA

Project Title
TANDATANI — A Verifiable Plot Passport for Every Smallholder

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Team Category	International (All Members)

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1. Executive Summary

TANDATANI (from Bahasa Indonesia *tanda* “mark” + *tani* “farmer” — an echo of *tanda tangan*, a signature) gives every smallholder plot a **cryptographically verifiable Plot Passport that the farmer owns**, so Indonesia’s 1.6 million coffee-farming families¹ — and after them the palm, cocoa and rubber smallholders behind up to **US\$7 billion** of exports² — are not locked out of Europe when the EU Deforestation Regulation takes effect on **30 December 2026**.^{3,4} The registration that doubles as a farmer’s geolocation record, e-STDB, covers just **0.03% of coffee farmers** today;⁵ TandaTani closes that gap at the price of a WhatsApp message.

A cooperative officer maps a plot in minutes on a **Bahasa-first, offline-capable app**; an **AI engine** cross-examines it against official satellite forest data (FAO Whisp, JRC-2020 baseline, GLAD/RADD alerts)^{6,7} and explains its verdict in plain Bahasa; the cooperative then signs a **W3C Verifiable Credential**⁸ the farmer keeps and can carry to *any* buyer, its evidence hash **anchored on the Polygon public chain** ($\approx$ \$0.001 per thousand plots⁹). Exporters click once for an **EU-ready due-diligence dossier** (TRACES GeoJSON)¹⁰ — and **anyone, including the judges reading this, can scan a QR and verify the entire chain of proof with no account, no permission, and no trust in us**.

Farmers pay nothing; exporters pay per shipment — less than 6% of a single consultant-day at today’s €800–1,500 rates¹¹ — for compliance that carries a documented **+US\$50/tonne premium** (Vietnamese-robusta benchmark).¹² Hybrid Web2 + Web3 + AI, every component free-tier, and every feature scoped to run **live, end-to-end, within the 24-hour final**.

2. Problem Statement

The problem in one sentence: from 30 December 2026, every kilogram of coffee, palm oil, cocoa or rubber entering the EU must carry plot-level geolocation proof that it grew on land that was not deforested after 2020 — and the Indonesian smallholders who grow almost all of it have effectively none of the required records.

2.1 A regulatory wall, eleven weeks after this hackathon’s final

Regulation (EU) 2023/1115 (EUDR) obliges any operator placing seven commodities on the EU market to file a due-diligence statement containing the **geolocation of every plot of origin** — a GPS point for plots up to 4 ha, a full polygon above that — plus evidence of **no deforestation after 31 Dec 2020** and of **legality** under Indonesian law.^{3,10} After two postponements, enforcement begins **30 December 2026** for large and medium operators (30 June 2027 for micro/small);⁴ the runway has been consumed, and the simplifications that came with the delay did not remove the plot-level data requirement. Non-compliance risks fines whose ceiling is no lower than **4% of a company’s EU-wide turnover**, plus confiscation and exclusion from public procurement.³ Indonesia is benchmarked “**standard risk**” — 3% of operators face ex-post audits, but a valid DDS reference is a customs precondition for *every* shipment, and buyers facing the fines push the data burden upstream regardless.¹³

2.2 The people on the wrong side of the wall

99.3% of Indonesian coffee production comes from smallholders on 0.5–2 ha plots ^{1,14}	0.03% of coffee farmers hold e-STDB — the registration that doubles as their geolocation record (MoA, 2024) ⁵	US\$7 B of Indonesian exports exposed to EUDR requirements (CMEA estimate) ²
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Indonesia is the world’s 4th-largest coffee producer (813,345 t in 2024)¹⁵, and its coffee is a smallholder crop to a degree few realize: **1.63 million farming households** work 98.1% of the coffee area.¹ The pattern repeats across the EUDR basket: smallholders hold **40.6% of the 16.8 M ha of oil palm** (≈2.4 M households),^{16,17} grow **99.8% of cocoa**¹⁸ and **≈91% of natural rubber**.¹⁹ Yet the state instrument that would prove their plots’ location and legality — the e-STDB cultivation registration — has reached **1.07% of palm farmers, 0.03% of rubber and coffee farmers, and 0.00% of cocoa farmers** (Ministry of Agriculture data, July 2024 — the latest published breakdown).⁵ By December 2023 exactly **15,054** e-STDB certificates existed against ≈2.4 million oil-palm households alone.¹⁷ Koltiva — the largest traceability provider in the country — itself estimates that **only ≈1% of Indonesian smallholders meet EUDR traceability and legality requirements**.²⁰

Field reality, August 2026

In Pasaman, West Sumatra, Minangkabau families grow coffee inside ancestral parak agroforests — shade trees, areca, sugar-palm and coffee layered “like a tidied forest.” It is the most forest-friendly coffee agriculture imaginable. But the plots “have no polygons and are not recorded in any database” — so, come 30 December, the most forest-friendly coffee has the highest chance of being refused at Europe’s door.²¹ Meanwhile in Gayo Lues, cooperative staff are walking member plots one by one, mapping by hand against protected-area boundaries.²²

2.3 Why the gap persists — it is not a mapping problem

A GPS point takes thirty seconds to capture. The real bottlenecks, documented across government and field studies, are structural:

- **Cost asymmetry.** Traditional certification routes cost **€86/ha upfront and €191–751 per farmer**, and can push a smallholder’s first-year net income **negative (–8%)**²³ — against typical plots of under 2 ha. Consultants bill **€800–1,500/day**.¹¹ These unit economics can never reach a 1-hectare farm.
- **Institutional throughput.** e-STDB issuance runs through district offices with limited capacity; the Agriculture Ministry has now opened data collection to third parties precisely because the state cannot map millions of plots alone.²⁴
- **A trust and custody gap.** Records collected by enterprise platforms belong to the paying client, not the farmer; Deputy Foreign Minister Havas Oegroseno put the farmer’s side bluntly — asked whether smallholders can perform EUDR’s screening duties: **“Can farmers do it? Of course not.”**²⁵



- **The digital divide is real but navigable.** Only **46.8%** of Indonesia's 28.2 M farmers use modern machinery or digital technology;²⁶ any workable solution must run offline, in Bahasa, on cheap Android phones — or through a literate cooperative officer.
- **Chain-of-custody fraud.** Collection points mix compliant and non-compliant lots; without tamper-evident records, one launderer can contaminate — legally and reputationally — an entire container.²⁷

2.4 The cost of doing nothing

The market is already pricing the wall in. EU-bound Indonesian coffee shipments jumped **+72% to 2.4 million bags** in 2025/26 as buyers front-load stock ahead of enforcement²⁸ — stockpiling, not confidence. Compliant robusta already earns a **+US\$50/tonne premium** in Vietnam, where 35–40% of area is EUDR-ready; non-compliant beans are not discounted into Europe — they are **excluded**, because mixing contaminates the lot.^{12,27} Physical segregation adds up to **+25%** logistics cost.²⁷ At stake for Indonesia: up to **US\$7 B of exports** by the government's own estimate,² at the very moment the newly concluded EU–Indonesia CEPA (Sept 2025) finally opens the €28.9 B trade corridor wider.²⁹ The door to Europe is being unlocked and walled up in the same year — and whether a farming family stays inside is decided by data they do not have.

Synthesis. Indonesia does not have a deforestation-evidence problem so much as a **proof-distribution** problem: the forests were standing in 2020, the satellites saw them, the farmers were there — but no instrument turns that truth into a portable, trusted, affordable record in a smallholder's hand. That is precisely an infrastructure gap — and infrastructure is what we build.

3. Proposed Solution & Value Proposition

Indonesia's harvest is real, the land is clean, but the evidence lives in paper folders, WhatsApp photos and enterprise databases the farmer will never control. TandaTani's answer: give the proof itself to the person who created it. Every ingredient we use exists; the **combination** — a compliance credential the *farmer* owns, priced *per shipment*, verifiable by *anyone* without an account — exists nowhere in production, and §3.5 shows why incumbents cannot copy it without breaking their own business model.

3.1 The core concept — a Plot Passport, owned by the farmer

For every plot we assemble one **evidence bundle**: geolocation (a GPS point for plots ≤ 4 ha, a walked polygon above — exactly the EU specification¹⁰), the satellite deforestation verdict, legality documents, geotagged field photos and the review trail. After human review, the cooperative signs this bundle as a **W3C Verifiable Credential 2.0**⁸ — the *Plot Passport*. Three properties make it more than a database row:

- **Portable.** The passport downloads as a self-contained signed file — the credential, its evidence hashes and its Merkle proof, plus a printable QR card — **held by the farmer** on their own device, with cooperative custody as the fallback, never locked in our database. Verifying it needs only that file and the public chain: if the farmer changes buyer or platform, or if TandaTani itself disappears, the proof still verifies. None of the major incumbents offers this; their records live in the paying client's silo.³⁰
- **Independently verifiable.** Each bundle's SHA-256 hash is batched into a Merkle tree whose root is anchored on the Polygon public chain.⁹ Any importer or auditor can re-



compute the hash in a browser and confirm the record existed, unaltered, at issuance — without asking us, or trusting us.

- **Private by design.** The chain carries only salted hashes — never coordinates, never names. Geolocation is disclosed solely inside the exporter’s due-diligence submission, as the regulation requires.³

Scope honesty — the legality pillar. EUDR demands two proofs: deforestation-free *and* legal production. TandaTani verifies the first from satellites; for the second it structures and hash-seals the farmer’s legality evidence (e-STDB, land letters, cooperative membership), checks it for presence and internal consistency, and records who attested it — while authenticity verification remains with the issuing authorities. The dossier is evidence the exporter files under its own due-diligence responsibility; we do not replace that duty — we make it affordable to discharge.

3.2 How it works — one journey

Ibu Sari, a field officer of a Gayo cooperative, visits member farms with a mid-range Android. At Pak Karim’s plot she opens **Ladang**, our farmer app: one tap for the GPS point, two photos, a harvest estimate — all offline; it syncs at the next signal. Overnight, the **AI engine** interrogates FAO Whisp and Global Forest Watch: was this forest on 31 Dec 2020? Any loss alerts since?^{6,7} PostGIS checks no other member has claimed an overlapping polygon. The verdict returns in Bahasa: “*Lahan ini aman — tidak ada deforestasi terdeteksi setelah 2020,*” with every evidence layer attached. The cooperative chair reviews her queue, taps **Terbitkan** (“issue”), and the cooperative’s key signs Pak Karim’s Plot Passport. When an exporter builds its next container for Hamburg, it selects verified passports and TandaTani emits the TRACES-format GeoJSON dossier¹⁰ plus a QR-coded evidence certificate. The importer scans the QR: hash matches, signature valid, root on-chain — due diligence the importer can re-verify, byte for byte, five years from now.

3.3 Why Web3 — an honest answer, not a buzzword

- **The trust gap is structural.** An EU importer must rely on evidence from a smallholder they will never meet, passed through 3–5 intermediaries — some of whom profit from laundering non-compliant volume into compliant lots.²⁷ A vendor’s private database asks the importer to trust the vendor and every editor; a public anchor makes tampering *detectable by anyone* — a signature proves *who* wrote a record, the anchor proves *when*, so no issuer can quietly backdate a passport after a deforestation alert lands.
- **Records must outlive companies.** EUDR requires operators to keep due-diligence records for **five years**.³ Startups die; anchored hashes on a public chain do not.
- **Portability breaks lock-in.** Verifiable Credentials give farmers *exit rights* over their own compliance history — the structural opposite of enterprise traceability, and the reason a farmer-first platform can win cooperatives’ participation.
- **Decentralized authentication where it belongs.** Exporters sign in with their wallets (SIWE); cooperatives hold signing keys (DIDs); farmers never touch crypto — a relayer pays the sub-cent gas. Web3 machinery, zero Web3 friction.

3.4 Why AI — verification at a price a smallholder can afford

Manual verification costs €800–1,500 per consultant-day;¹¹ corporate mapping programs have burned millions. TandaTani’s engine performs the equivalent convergence-of-evidence analysis in seconds for **≈ \$0.002 per plot**: deterministic geospatial scoring over official



datasets (JRC forest baseline 2020, GLAD/RADD alerts, commodity maps via Whisp, with Global Forest Watch as an independent backup track)^{6,7} plus PostGIS geometry-anomaly detection against duplicate or overlapping claims — then **Gemini** (official Bahasa support³¹) renders the verdict as three plain sentences a farmer can act on, and powers **Tanya Tani**, a field-question copilot. The design principle is auditable honesty: **AI narrates; evidence decides**. Every verdict links its raw sources.

3.5 The landscape — and the wedge it leaves open

Today's options	Who pays, roughly	The gap TandaTani exploits
Enterprise traceability — Koltiva, osapiens, Sourcemap, Agridence ³⁰	Exporters/brands: €10k–50k+/yr + €5k–90k setup ¹¹	Top-down; the farmer is a data row; records sit in the client's silo
Web3 agri-platforms — Dimitra × PT Surveyor Indonesia ³²	Enterprise deals; fees tied to the DMTR token	Token volatility as a compliance dependency; credentials not farmer-owned
NGO transparency — Fairfood Trace (Hedera) ³³	Donor-funded pilots	Premium storytelling, not EUDR dossier automation; no AI verification
Consultants & certifiers ^{11,23}	€800–1,500/day; €191–751 per certified farmer	Unit economics that can never reach a 2-ha farm
Free public tools — Whisp, GFW, TRACES, Fairtrade Plot Insights ^{6,34}	Free	Analysis and filing exist — identity, attestations, volume-bounded dossiers, offline Bahasa capture and portable credentials do not

Unique selling proposition. TandaTani is the only design in this landscape where the *farmer* ends up owning a portable, publicly verifiable compliance credential — free — while the exporter pays per shipment instead of per year. Incumbents cannot copy this without dismantling their enterprise pricing and data custody; that structural conflict, not secrecy, is our moat. And we are deliberately **complementary to the state**: Indonesia's blockchain-based National Dashboard and the e-STDB drive need exactly the verified, plot-level smallholder data we produce — the Agriculture Ministry now invites third parties to collect it.^{24,35} TandaTani is the missing last mile between a farmer's field and every rail built above it.

4. Minimum Viable Product (MVP) / Key Features

Everything below will be **built and running end-to-end during the 24-hour final** — live backend, live satellite queries, live chain writes, on a public URL. Nothing hardcoded, nothing mocked. Scope discipline is the strategy: five features, each independently demoable, composing one continuous story.

Feature 1 — Ladang: offline-first plot capture (Bahasa)

A PWA for low-end Android. One-tap GPS point for plots ≤ 4 ha (per the EU geolocation spec;¹⁰ smartphone GPS reaches ≤ 4.9 m in open sky³⁶, and under shade canopy Ladang shows live accuracy, accepting a point only below a 10 m threshold with tap-on-imagery fallback) or walk-the-boundary polygon capture above 4 ha, over Esri satellite imagery (MapLibre GL). Farmer profile, plot photos, harvest estimate. Captures queue in IndexedDB and sync when signal returns — demonstrated live in airplane mode.





Feature 2 — AI verification engine

On sync, the backend submits each geometry to **FAO Whisp**⁶ (convergence of evidence across JRC forest-cover 2020, tropical-moist-forest, GLAD-L/S2 and RADD alerts, ESA WorldCover, commodity maps) with the **Global Forest Watch API** as an independent second track.⁷ PostGIS runs ST_Intersects duplicate/overlap detection against every existing plot. Results fuse into a deterministic risk score; **Gemini 2.5 Flash**³¹ renders the verdict as three plain-Bahasa sentences with evidence links. The demo shows a clean plot passing and a test polygon drawn over post-2020 forest loss failing — live.

Feature 3 — Plot Passport issuance (Cooperative Console)

A review queue for the cooperative: map, AI verdict, photos and anomaly flags side by side; approve or reject in one tap. On approval the platform issues a **W3C Verifiable Credential 2.0**, Ed25519-signed by the cooperative’s did:key.⁸ The passport — with its QR — appears instantly in the farmer’s app.

Feature 4 — Public anchoring & one-scan verification

Evidence hashes batch into a Merkle tree; the root is anchored by **TandaRegistry.sol** on Polygon (Amoy testnet in the demo; the identical call costs ≈ \$0.001 on mainnet⁹). The public verify page re-computes the bundle hash in-browser and checks the VC signature plus the Merkle proof against the on-chain root — no login, no API key, no trust in TandaTani.

Judges can scan the QR from their own phones during the pitch.

Feature 5 — Exporter DDS Studio

Wallet sign-in (SIWE); pick verified passports into a batch; export a due-diligence dossier — **TRACES-conformant GeoJSON** (WGS-84, six-decimal precision, point-or-polygon by plot size¹⁰) plus a QR-stamped evidence-summary PDF. A **volume-plausibility gate** blocks any dossier whose tonnage exceeds the summed harvest estimates of its passports — the launderer’s math stops working — and an append-only audit log shows every state change behind it.

The 24-hour execution plan (3 builders × 24 h)

Window (WIB)	Builder 1 (S. Gupta) — Web3 / Infra	Builder 2 — Frontend / PWA	Builder 3 — Backend / AI
10:00–14:00	Repo/CI; Vercel + Supabase; deploy TandaRegistry.sol to Amoy	App shell; map capture; offline queue	Supabase schema + PostGIS; API skeleton
14:00–19:00	VC issuance service (did:key, Ed25519)	Farmer flows; verification result UI	Whisp + GFW integration; risk scorer
19:00–00:00	Merkle batcher + anchoring worker	Cooperative Console	Gemini Bahasa verdicts; Tanya Tani
00:00–05:00	Public verify page (hash + proof in browser)	Exporter DDS Studio	TRACES GeoJSON + PDF dossier generator
05:00–08:00	Integration pass, all hands — seeded Gayo demo dataset, airplane-mode drill, cross-device tests		
08:00–10:00	Hardening, deploy freeze, pitch rehearsal ×3, Q&A drill		

Feasibility rests on rehearsed pieces, not hope. Every external service (Whisp, GFW, Gemini, Amoy faucet, Supabase, Vercel) is free-tier, with accounts and API keys provisioned before the clock starts. Fallbacks are pre-planned: GFW if Whisp queues slowly; Groq/Llama behind the same interface if Gemini rate-limits; a cached verification fixture exists *only* as an offline-

resilience aid and would be labeled as such on screen, never presented as live output. **Cut-not-fake rule: if a feature slips, we cut it — we never simulate it.** Declared cut order if time slips: the Tanya Tani copilot goes first, the PDF dossier second (the TRACES GeoJSON remains), Exporter Studio polish third — the capture → verify → issue → publicly-verify loop (Features 1–4) is the non-negotiable demo core.

Deliberately out of the 24-hour scope — native mobile builds, payments, marketplace, mainnet deployment, SD-JWT selective disclosure, voice interface. They appear in §5 as roadmap and are promised nowhere as demo.

5. Target Market & Impact Viability

5.1 Exactly who uses TandaTani

- **Primary — cooperative field officers and managers** in Indonesia’s Arabica belt: specifically the Gayo highlands (Aceh Tengah, Bener Meriah, Gayo Lues), home to Southeast Asia’s largest Arabica area, a protected Geographical Indication origin since 2010, and cooperatives like women-led **Kopepi Ketiara (2,000+ members across 19 villages)** — which face a **January 2027** Fairtrade geolocation deadline on top of EUDR.^{34,37} These are digitally literate professionals mapping member plots *by hand* today.²²
- **Primary — compliance managers at Indonesian coffee exporters** shipping green beans to Belgium and Germany (the top two EU destinations²⁸), who currently face consultant day-rates or enterprise SaaS to produce every due-diligence statement.¹¹
- **Secondary (year 1–2)** — cocoa and rubber cooperatives (0.00% and 0.03% e-STDB coverage⁵), then independent palm smallholder associations; **EU importers** who verify passports (free) and later pay for API access.
- **Validation, honestly stated:** we have not yet interviewed a customer. Between this submission and the final we will hold structured calls with Gayo cooperative officers (intro path: the WRI Sustainable Gayo Coffee network and Fairtrade producer support) to test willingness-to-pay — and we will report what we hear, including bad news, on stage.

5.2 Market size, from the bottom up

Scope	Basis	Size
TAM — EUDR due-diligence software, global	Analyst estimate, 2026 → 2036 ³⁸	US\$1.1 B → 1.8 B
SAM — Indonesian smallholder EUDR corridor	≈7.5 M smallholder households across coffee (1.63 M), palm (2.4 M), cocoa (1.4 M), rubber (2.1 M); exports at stake up to US\$7 B ²	≈7.5 M farms · US\$7 B trade
SOM — 12-month beachhead: Gayo coffee	(8,000 members × €1.40 × 2 seasons) + (200 dossiers × €45) ≈ €31k, across 3 co-ops and 2 exporters	≈US\$33k ARR

5.3 Business model — commercial from day one

- **Farmers: free, forever.** Their participation is the network; charging them kills it.
- **Exporters: Rp 800k (≈ €45) per due-diligence dossier**, or a flat coop-season license. Context: consultants run €800–1,500/day, enterprise SaaS €10k–50k/yr;¹¹ one compliant

19-tonne container carries $\approx$ US\$950 of premium at +US\$50/tonne.¹² We price at $\approx$ 5% of the value we unlock.

- **Cooperatives: Rp 25k ($\approx$ €1.40) per member per season** for the credential registry and Fairtrade/e-STDB exports — less than 1% of a certification's per-farmer cost;²³ a member visit takes minutes, and one visit now serves EUDR, Fairtrade and e-STDB at once.
- **Later: verification API** for EU importers and (per Ditjenbun's invitation to third parties²⁴) **paid e-STDB data-collection partnerships** feeding the National Dashboard.³⁵

Unit economics. Marginal cost per verified plot $\approx$ \$0.002 LLM + anchoring amortized to $\approx$ \$0.000001 in 1,000-plot batches + fractions of a cent of storage^{9,31} — under **\$0.01 all-in** against €45-per-dossier revenue: software margins on infrastructure that incumbents deliver with field armies. Free tiers carry the pilot; the first paying exporter carries the company.

5.4 Scalability — one engine, many walls

The pipeline — *map* $\rightarrow$ *verify* $\rightarrow$ *credential* $\rightarrow$ *anchor* $\rightarrow$ *dossier* — is commodity-agnostic and regulation-agnostic. The same engine serves cocoa, rubber and palm; the same passports satisfy Fairtrade's 2027 geodata mandate,³⁴ feed e-STDB and the National Dashboard,³⁵ and are ready for whatever follows EUDR (the UK and US have parallel bills in motion). Geographically, Vietnam alone has 60–65% of its coffee area still non-compliant;¹² the architecture ports with a language file. Because passports are portable open credentials, every new buyer that accepts them makes every existing passport more valuable — adoption compounds.

5.5 Impact, measured

- **Economic:** keeping smallholders inside a corridor worth up to US\$7 B² and capturing the +US\$50/t compliance premium¹² at the farm gate rather than the trader's margin.
- **Institutional:** every passport is e-STDB-ready data — accelerating the registration drive the government itself says it cannot staff.²⁴
- **Environmental:** EUDR's promise only works if compliance is achievable; TandaTani makes deforestation-free farming *provable*, so the market can reward it (SDGs 1, 8, 12, 13, 15).
- **Social:** the first bankable, portable proof-of-practice a smallholder has ever owned — with women-led cooperatives like Ketiaras³⁷ as first protagonists.

5.6 Risks, named — and answered

Risk	Reality check	Mitigation built into the design
Farmer digital literacy (46.8% modern-tech adoption ²⁶)	Real, generational	Cooperative-officer-assisted capture; offline PWA; Bahasa-first; voice roadmap
EUDR slips a third time	Possible — it slipped twice ⁴	The 2024 delay itself triggered +72% buyer front-loading — buyers, not Brussels, now enforce; ²⁸ passports also serve Fairtrade Jan-2027, e-STDB and ISPO ^{24,34}
GPS spoofing / claim fraud	Attempted everywhere	Satellite cross-check, PostGIS overlap detection, geotagged photos, cooperative attestation — convergence, not trust
Cooperative key loss	Operational hazard	Credential-status registry + re-issuance under a successor DID (did:key is single-keypair by design; roadmap: did:web for true rotation)

Incumbent copies the model	Koltiva has 1.9 M producers ³⁰	Copying farmer-owned portability breaks their enterprise custody model — our wedge is their conflict
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6. Tech Stack & System Architecture

One hybrid system, three planes: a **Web2 core** for speed and cost, a **Web3 trust plane** for proof and portability, an **AI plane** for verification at smallholder prices. Every component is free-tier; the whole demo runs on public URLs.

6.1 Data flow (the numbered steps match the chips in Figure 1, next page)

1. Field capture, fully offline: GPS point/polygon, photos, farmer identity → IndexedDB queue.
2. Sync to the API (Next.js route handlers, Zod-validated); plots land in Postgres/PostGIS,³⁹ where the ST_Intersects gate blocks exact-duplicate claims outright and flags partial overlaps into the risk score.
3. AI engine gathers satellite evidence (Whisp primary; GFW independent track)^{6,7} and Gemini writes the Bahasa verdict.³¹
4. Cooperative reviews the AI-annotated queue in its Console.
5. Approved bundle → W3C Verifiable Credential 2.0, signed with the cooperative's did:key (Ed25519).⁸
6. Bundle hashes batch into a Merkle tree — thousands of plots, one 32-byte root.
7. The root is anchored by TandaRegistry.sol on Polygon; the event is public forever.⁹
8. Exporter composes verified passports → TRACES GeoJSON + evidence-PDF dossier.¹⁰
9. Anyone scans the QR: the browser re-computes the hash and checks VC signature + Merkle proof against the chain.

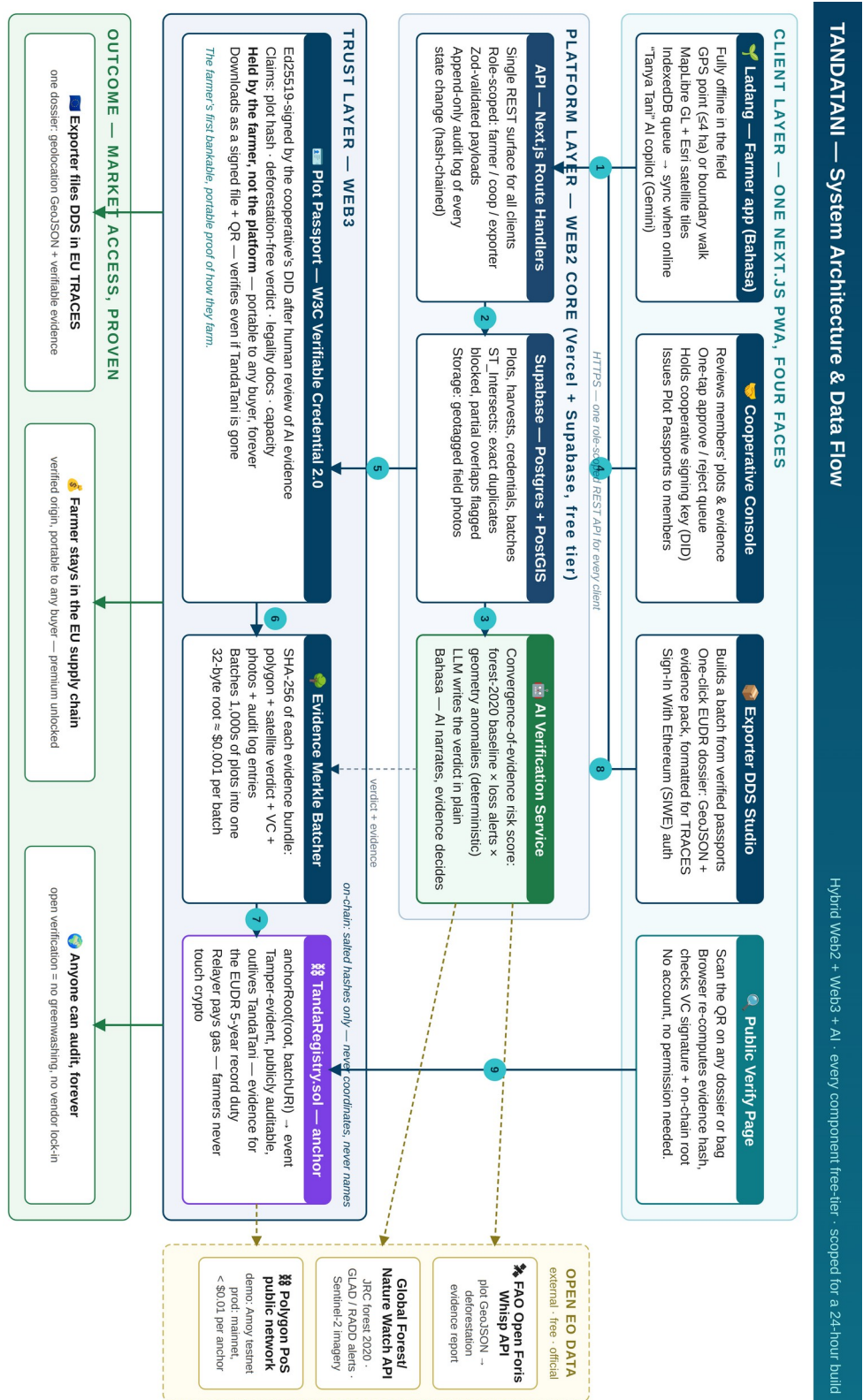


Figure 1 — TandaTani system architecture and data flow (rotate to read). Numbered chips on the arrows match the data-flow steps in §6.1.



6.2 Technology choices — and why each earns its place

Layer	Technology	Why this, for this build
Frontend	Next.js 15 PWA · TypeScript · Tailwind · MapLibre GL + Esri World Imagery · Dexie (IndexedDB)	One codebase, four roles; installable on low-end Android; offline by architecture, not afterthought
Backend	Next.js route handlers on Vercel · Supabase: Postgres + PostGIS, Auth, Storage ³⁹	Zero-ops free tiers; geometry fraud checks are one SQL clause
AI / EO	FAO Whisp API · Global Forest Watch Data API · Gemini 2.5 Flash, Groq/Llama fallback ^{6,7,31}	Official, free, EUDR-aligned convergence of evidence; Bahasa officially supported
Web3	Solidity (Foundry) on Polygon PoS/Amoy · viem · @digitalbazaar/vc + did:key (Ed25519) · SIWE · OpenZeppelin Merkle ^{8,9}	Sub-cent anchoring; W3C-standard credentials (Recommendation, May 2025); wallets only where users are companies
Delivery	GitHub Actions CI · Vercel previews · public demo URL + Amoy explorer links	Live public deployment for the pitch — judged explicitly above localhost

6.3 The on-chain / off-chain boundary (privacy by architecture)

- **On-chain:** 32-byte Merkle roots and batch URIs — nothing else. $\approx$ \$0.001 per thousand plots⁹: the whole Gayo highlands could be anchored for the price of one cup of its own coffee.
- **Off-chain:** coordinates, identities, photos — Postgres under row-level security, disclosed only inside the exporter's DDS as EUDR requires.³ Salted hashing defeats rainbow-matching of known plots.
- **Keys:** cooperatives hold did:key signing keys; exporters authenticate by wallet (SIWE); a relay pays gas so farmers never see crypto.
- **Privacy law, named.** Indonesia's PDP Law (UU 27/2022) and the GDPR shape the design: consent captured at enrollment (the cooperative as data controller, TandaTani as processor), coordinates under row-level security and disclosed only inside the DDS, credential copies on the farmer's device and printable QR card — and, because the chain holds only salted hashes, destroying a record's salt is a clean erasure path. The public QR page shows a verdict, never coordinates or identities.

6.4 Engineering honesty — failure modes we designed for

- Whisp jobs are asynchronous (Earth-Engine-backed): we submit early, poll, and keep GFW as an independent evidence track — either alone yields a verdict.^{6,7}
- Background Sync API is Chromium-only, so the offline queue flushes on the browser's online event — works everywhere modern; demoed on Android Chrome.
- Free-tier LLM limits (Gemini: 250 requests/day³¹) comfortably exceed rehearsal-measured demo load; Groq is wired behind the same interface.
- If venue Wi-Fi dies mid-pitch, the offline-first design *is* the contingency: capture continues, and we sync over any hotspot.

References

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